

11/23/23, 12:50 AM

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embedding.py x

```
1 import os
2 import openai
3 import secret
4 openai.api_key=secret.api_key
5
6 # WRITE YOUR CODE HERE
7 emb=openai.Embedding.create(
8     model="text-similarity-babbage-001",
9     input="You will rejoice to hear that no disaster has accompanied the")
10
11 print(len(emb['data'])[0]['embedding']))
```

100% (11:40)

https://codio.com/sandipan-dey/endpoints/tree/embedding.py

Codio - Endpoints

Guide x

Collapse

Embeddings

Embedding Endpoint

Embeddings Endpoint Signature

POST https://api.openai.com/v1/embeddings

The API also has another experimental endpoint called *embeddings*. Embeddings are a representation of a given input as a vector of floating point numbers. Embeddings can be easily consumed by machine learning models and algorithms. They are often used to determine the semantic similarity between two texts. If two texts are similar, then their vector representations should also be similar.

Currently OpenAI offers three families of embedding models that allow for text search, text similarity and code search. Each family includes up to four models on a spectrum of capabilities:

- **Ada** (1024 dimensions),
- **Babbage** (2048 dimensions),
- **Curie** (4096 dimensions),
- **Davinci** (12288 dimensions).

These embedding models are specifically created to be good at a particular task. For example, `text-similarity-ada-001` is good at capturing semantic similarity between two or more pieces of text. Or `text-search-ada-doc-001` helps measure whether long documents are relevant to a short search query. For more information on embeddings, see the OpenAI documentation.

Creating an Embedding

In this example, we are going to use the `text-similarity-babbage-001` model. We also need to provide some text for the embedding. This is done with the `input` keyword argument. Both of these are required. The `user` keyword argument is optional. This unique identifier can be used to help monitor for abuse. The code sample below should print a long list of floating point numbers.

```
emb=openai.Embedding.create(
    model="text-similarity-babbage-001",
    input="You will rejoice to hear that no disaster has accompanied the")

print(emb)
```

TRY IT!

2048

Just like the list of models, the return object is of type `OpenAIObject`. If you want to access just the vector, change the print statement to the following:

```
print(emb["data"][0]["embedding"])
```

TRY IT!

2048

Which of the following is true about embeddings?

☐ They are a type of machine learning model.

☐ They are a way to represent text as a sequence of integers.

☒ They are a type of vector representation of input data.

☐ They are only used for text search and text similarity.

They are a type of vector representation of input data. They are often used in machine learning and natural language processing to convert categorical variables to continuous vectors. Embeddings essentially provide a way of reducing dimensionality and dealing with sparse data.

Check It!

Next